

Spike-Based Hand Gesture Recognition: A Comparative Study of Neuromorphic Encoding Strategies Using Izhikevich Neurons

Neuromorphic Engineering – Project Presentation

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Outline

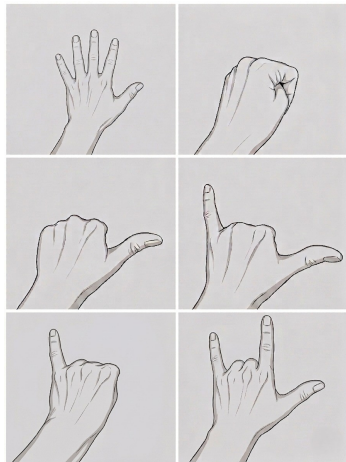
- 1 Introduction
- 2 System Architecture
- 3 Neuromorphic Encoding
- 4 Results
- 5 Standard vs. Neuromorphic Comparison
- 6 Conclusions & Future Work

Objective

Compare **standard geometric** and **neuromorphic spike-rate** approaches for classifying 6 left-hand gestures using the **Leap Motion Controller (LMC)**.

Gestures classified:

- 1 Open palm
- 2 Fist
- 3 Thumb only
- 4 Pinky only
- 5 Horns gesture
- 6 Thumb + Pinky



Leap Motion Controller

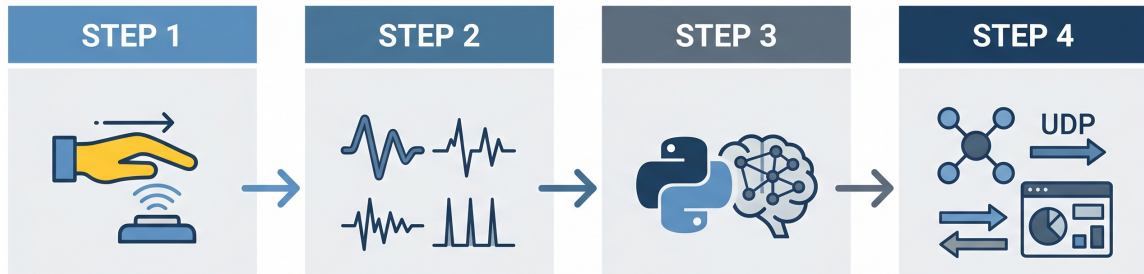
- 2 monochromatic IR cameras + 3 IR LEDs
- Tracks fingertip positions, palm position, hand direction
- **Limitation:** when a finger is not detected, the output matrix shrinks without channel discrimination

Key challenge

Ambiguous finger-channel assignment $\Rightarrow$ need for a spatial routing mechanism



Pipeline Overview: LabVIEW + Python



LabVIEW

Acquisition, routing,
Izhikevich conversion

Python

RF training, Grid Search,
.pkl serialization

LabVIEW (RT)

Live inference via
Python Node + UDP

LabVIEW Spiking Framework Architecture



- **VI 1: Data-to-Spike Encoder**
Used for offline processing to transform static raw datasets into temporal spike trains.
- **VI 2: Real-Time Sender**
Performs live signal acquisition and conversion into spikes.
- **VI 3: Inference Receiver**
Processes the incoming spike stream to perform complex **regression** and **classification** of hand gestures.

Random Forest + Grid Search

Training setup

- **Classifier:** Random Forest
- **Split:** 75% train / 25% test
- **CV:** 5-Fold Stratified

Grid Search space (36 combos)

- `n_estimators`: 50, 100, 200
- `max_depth`: None, 10, 20
- `min_samples_split`: 2, 5
- `min_samples_leaf`: 1, 2



The Izhikevich Neuron Model – Equations

Two differential equations that reproduce a wide range of spike patterns:

$$\begin{aligned}\dot{v} &= 0.04v^2 + 5v + 140 - u + I \\ \dot{u} &= a(bv - u)\end{aligned}$$

reset rule applied after each spike ($v \geq 30$ mV):

$$\text{if } v \geq 30 \text{ mV : } \begin{cases} v \leftarrow c \\ u \leftarrow u + d \end{cases}$$

In this project:

- $I_{\text{Finger}} = \text{Distance}(\text{Tip-Palm}) / 10$
- $I_{\text{Direction}} = \text{Normalized}[3.8 - 11]$
- 5 to 11 Neurons implemented

Variables

- v : membrane potential [mV]
- u : membrane recovery variable
- I : input current (encodes finger distance)

Parameters (a, b, c, d)

- a : time scale of recovery u
(small $a \Rightarrow$ slow recovery)
- b : sensitivity of u to v
(larger $b \Rightarrow$ stronger coupling)
- c : after-spike reset value of v
(more negative $\Rightarrow$ deeper reset)
- d : after-spike increment of u
(larger $d \Rightarrow$ stronger adaptation)

Neuron Types & Rate Coding

Rate Coding Principles:

- **Spike-count rate coding** maps a physical signal's amplitude onto the neuron's firing frequency:

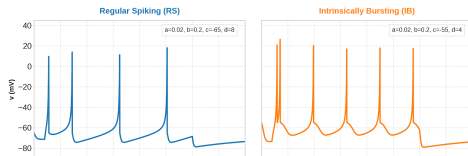
$$r = \frac{N_{\text{spikes}}}{T_{\text{window}}}$$

- Higher distance $\Rightarrow$ higher $I \Rightarrow$ higher firing rate.
- Window $T_{\text{window}} = 1 \text{ s}$ in this project.
- Feature used: **spike count** N_{spikes} (ISI discarded).

Different (a, b, c, d) combinations reproduce distinct **electrophysiological behaviours**.

Izhikevich Neuron Models

Type	a	b	c	d
Regular Spiking	0.02	0.2	-65	8
Fast Spiking	0.1	0.2	-65	2
Chattering	0.02	0.2	-50	2
Low-Threshold	0.02	0.25	-65	2
Resonator	0.1	0.26	-65	2
Intrins. Bursting	0.02	0.2	-55	4
Thalamo-cortical 1	0.02	0.25	-65	0.05
Thalamo-cortical 2	0.02	0.2	-65	8



Different approaches & encoding strategies

Configuration	Channels	Notes
Blind Spike Encoding	5	raw coordinates, no routing
Summed Spike Encoding	1	sum of all channel spikes
Routed Spike Encoding	5	x-axis normalized, per-finger
BlindDir Spike Encoding	11	raw + palm direction ($\pm$ splits)
Distances Only (standard)	5	raw euclidean palm - tip

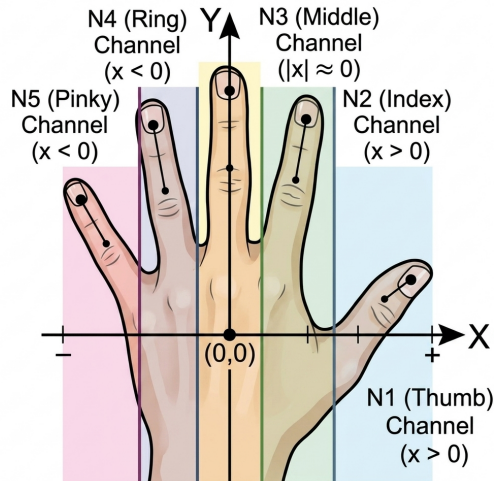
Spatial Routing Mechanism

Relative x-coordinate for each finger i :

$$x_{rel,i} = x_{finger,i} - x_{palm}$$

- **Middle (N3):** $|x_{rel,3}| \approx 0$
- **Ring (N4), Pinky (N5):** $x_{rel} < 0$
- **Index (N2), Thumb (N1):** $x_{rel} > 0$

Fixed thresholds partition the 1D lateral space into 5 finger channels, solving the ambiguous-channel problem.

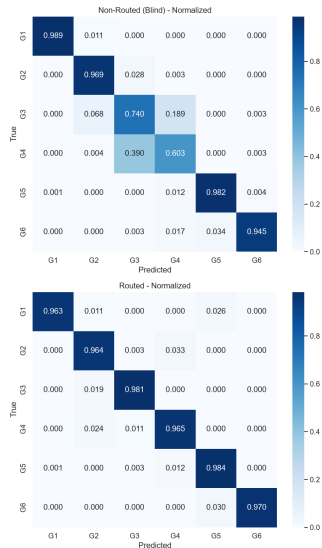


Neuromorphic Results: Routing vs No-Routing

Neuron Type	With Routing	Without Routing
Fast	0.9754 $\pm$ 0.0113	0.8561 $\pm$ 0.0233
Chattering	0.9716 $\pm$ 0.0168	0.8637 $\pm$ 0.0190
Low-Threshold	0.9716 $\pm$ 0.0146	0.8694 $\pm$ 0.0273
Regular	0.9716 $\pm$ 0.0133	0.8713 $\pm$ 0.0160
Resonator	0.9735 $\pm$ 0.0110	0.8845 $\pm$ 0.0067
Thalamo2	0.9716 $\pm$ 0.0133	0.8713 $\pm$ 0.0160
IntrinsicallyB	0.9678 $\pm$ 0.0128	0.8542 $\pm$ 0.0207
Thalamo1	0.9640 $\pm$ 0.0109	0.8675 $\pm$ 0.0099

Table: Model Accuracy Comparison.

- Routing yields +8.9% to +11.7% accuracy boost
- **Fast + Routing** = best neuromorphic: 97.54%

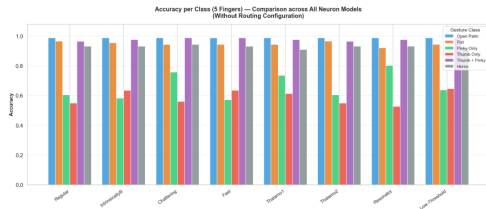


Class-wise Accuracy Without Routing

Gesture	Fast	Resonator
Open Palm (G1)	0.9885	0.9885
Fist (G2)	0.9535	0.9651
Thumb Only (G3)	0.6444	0.8667
Pinky Only (G4)	0.6667	0.5806
Horns (G5)	0.9765	0.9765
Thumb+Pinky (G6)	0.9655	0.9425

Critical Observation

Single-finger gestures (G3, G4) are **severely confused** without routing — same active-channel count, similar distances.



Spike Sum Configuration (1 Channel)

- Spikes from all 5 fingers are summed together
- **Reduce** multi-dimensional inputs into a single scalar
- Minimal computational complexity

Best result: Resonator Neuron

83.52% $\pm$ 2.90% (5-fold CV)

Highest accuracy for 1-channel configuration

Neuron	Accuracy (%)
Resonator	83.52 $\pm$ 2.90
Chattering	82.02 $\pm$ 2.95
Regular	81.07 $\pm$ 1.74
Thalamo2	81.70 $\pm$ 1.74
Fast	80.30 $\pm$ 0.78
Low-Threshold	80.88 $\pm$ 2.29
IntrinsicallyB	77.85 $\pm$ 1.67
Thalamo1	78.98 $\pm$ 1.65

Confusion Matrix — Blind Sum (Resonator Neuron)
Accuracy: 83.33% (5-Fold CV)

	Open Palm	Fist	Pinky Only	Thumb Only	Thumb + Pinky	Horns
Open Palm	66 (95.9%)	1 (1.1%)	0	0	0	0
Fist	0	79 (91.2%)	5 (5.8%)	2 (2.3%)	0	0
Pinky Only	0	8 (8.8%)	69 (76.7%)	13 (14.4%)	0	0
Thumb Only	0	2 (2.3%)	47 (92.6%)	29 (81.9%)	0	5 (5.4%)
Thumb + Pinky	0	0	8 (1.2%)	84 (98.8%)	0	0
Horns	0	0	0	3 (3.4%)	1 (1.1%)	83 (95.4%)

True Label

Open Palm Fist Pinky Only Thumb Only Thumb + Pinky Horns

Predicted Label

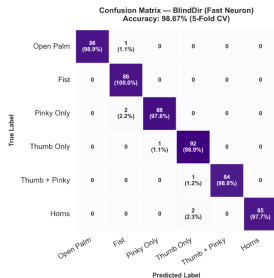
BlindDir Configuration (11 Channels)

- 5 finger spike channels (raw)
- 6 direction channels
($\pm$ split for each of X, Y, Z)
- Preserves sign without spatial normalization
- Exactly 3 direction channels active

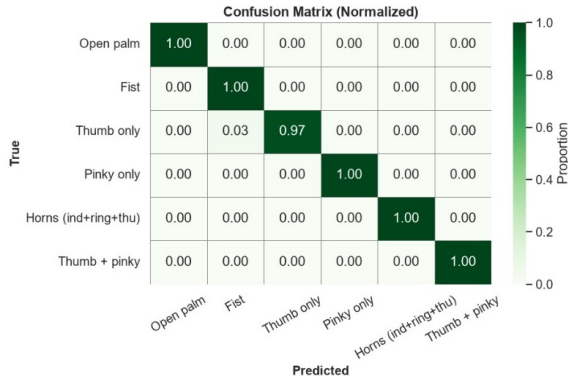
Best result: Fast Neuron

98.49% $\pm$ 0.96% (5-fold CV)
Highest accuracy across all configurations

Neuron	Accuracy (%)
Fast	98.49 $\pm$ 0.96
Thalamo1	98.48 $\pm$ 0.97
Chattering	98.11 $\pm$ 1.34
IntrinsicallyB	97.92 $\pm$ 1.62
Regular	98.10 $\pm$ 1.04
Low-Threshold	97.92 $\pm$ 1.25
Resonator	97.73 $\pm$ 1.53
Thalamo2	97.72 $\pm$ 1.62



Standard Results



- The direction, in this case, is useless.
- **+10%** higher accuracy compared to the Neuromorphic approach.
- **NOTE:** This approach was not tested and implemented in LabVIEW.

Configuration	Accuracy (%)
Distances Only (5 feat.)	99.32

Information Representation Trade-off

Configuration	Channels	Preprocessing	Fast Acc.
Routed	5	Spatial normalization (float ops)	97.54%
BlindDir	11	None (direct spike encoding)	98.67%
Standard	5	Raw euclidean distances	99%

Engineering insight

Routed transmits **less than half** the data of BlindDir, while keeping accuracy virtually unchanged — ideal for ultra-low-power neuromorphic hardware.

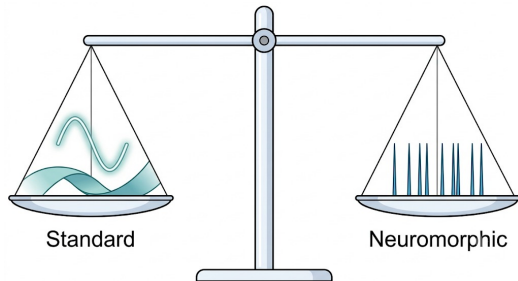
Information Representation Trade-off

Standard Approach

- 32/64-bit floats
- High precision
- Distances alone → **99% accuracy**
- Heavy dataset.

Neuromorphic Approach

- Sparse binary spikes, rate-coded
- Tiny integer counts per window
- BlindDir Fast → **98.67%** (only $\approx 1.33\%$ drop)
- High dataset reduction



Temporal Window Optimization: Blind Resonator Case Study

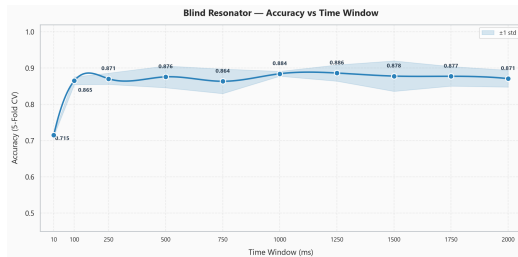
Impact of T_{window} on Classification

Performance:

- **Small time window:** Accuracy is extremely low (71.5%).
- **Plateau and Saturation (100 ms – 2000 ms):** Performance stabilizes, reaching a maximum of 88.6% at 1250 ms..

Key Takeaway:

- Optimal window at 100 ms → keeping latency low without sacrificing classification accuracy.
- Only the 1000 ms window was tested, so its the one implemented.



Does it actually work in reality when communicating via UDP?

The transition from theoretical models to real-world application yields mixed results, which strongly depend on the chosen encoding strategy:

- **Blind approach:** **Works reliably.** It is robust enough to handle the transition to real-time UDP communication without major issues.
- **Blind + Dir & Routing approaches:** **Encounter practical issues.** Their performance is highly dependent on the specific measurement conditions during data acquisition. In practice, the user is forced to position and move their hand in a very strict and predefined manner relative to the Leap Motion sensor, making them less robust in a completely unconstrained real-time scenario.

Conclusions & Future Work

- **Spatial representation is critical:** Collapsing to a single spike-sum severely degrades accuracy; routing or direction vectors restore it.
- **Neuromorphic efficiency is compelling:** $\sim 1.33\%$ accuracy loss vs standard, in exchange for sparse, binary, event-driven representations — fully compatible with low-power edge hardware.

1. Robust routing mechanism

- Use other type of routing to avoid the high dependency of the position of the hand during the gesture

2. Dataset expansion

- More subjects, diverse hand anthropometries
- Verify robustness of fixed routing thresholds
- Validate biological spike encoders generalization

Thank you!

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